

Darukaa.Earth

Name: Sarang Patil

BE- AI&DS

SIES Graduate School Of Technology

1. Project Description

Darukaa.Earth is a comprehensive, full-stack web application designed for environmental conservation professionals to monitor, track, and manage nature-based solutions. The platform allows users to create projects (e.g., Reforestation, Wetland Conservation), draw geospatial polygons on interactive maps to designate specific sites, and visualize the environmental impact through detailed charts covering Carbon Sequestration and Biodiversity metrics.

The system features secure user authentication, responsive design, interactive Mapbox-powered geospatial tools, and dynamic data visualization to make environmental reporting seamless and accurate.

2. Technology Stack

The application is built using a modern, scalable, and robust architecture divided into a frontend and a backend.

Frontend

- Framework: React (Vite, JavaScript/JSX)
- Routing: React Router DOM
- Styling: Custom Vanilla CSS with CSS Variables (Design System)
- Mapping & Geospatial: Mapbox GL JS, @mapbox/mapbox-gl-draw
- Data Visualization: Chart.js, react-chartjs-2
- HTTP Client: Axios
- Icons: Lucide React

Backend

- Framework: FastAPI (Python)
- Database: PostgreSQL
- ORM & Migrations: SQLAlchemy 2.0, GeoAlchemy2, Alembic
- Authentication: JWT (JSON Web Tokens), Passlib (Bcrypt)
- Data Validation: Pydantic
- Server: Uvicorn

Infrastructure

- Containerization: Docker, Docker Compose
- Version Control: Git & GitHub

3. Important Links

GitHub Repository: [GitHub Link](#)

Demo Video: [Video Link](#)